

Bayesian Matrix Denoising: Variational Inference for the Matrix Laplace Prior via Gaussian Scale Mixtures

David A Knowles

Abstract

We present a Bayesian approach to low-rank matrix estimation under a heteroscedastic noise model. By utilizing a novel matrix Laplace distribution we encourage low-rank structure directly on the matrix X without resorting to a non-convex factorization. To overcome the non-differentiability and non-separability of the prior, we exploit its variational representation as a Gaussian Scale Mixture. This auxiliary variable formulation allows us to develop a Coordinate Ascent Variational Inference (CAVI) algorithm that is strictly convex in all sub-updates, naturally estimates the global regularization parameter, and avoids the Neyman-Scott paradox by explicitly incorporating posterior uncertainty.

1 Introduction

Low-rank matrix factorization is a workhorse of modern data analysis, underpinning methods ranging from collaborative filtering to genomics. Yet the overwhelming majority of factorization approaches including non-negative matrix factorization (NMF), independent component analysis (ICA), and sparse Principal Component Analysis (PCA), optimize a non-convex objective over a product of factor matrices $X = UV^T$. Non-convexity has practical consequences beyond mere theoretical unease: different random initializations routinely converge to qualitatively different local minima, making results difficult to reproduce and the recovered factor structure ambiguous [1, 2]. Classical PCA is the notable exception, enjoying a globally optimal and unique solution via the singular value decomposition, but it cannot handle missing data or heteroskedastic noise, and it provides no mechanism for selecting the rank of the solution.

Convex relaxations offer a principled escape from local minima. The nuclear norm $\|X\|_* = \sum_i \sigma_i(X)$, the convex envelope of matrix rank, yields a globally convex objective that implicitly encourages low-rank solutions without factorizing X [3, 4]. Solving the nuclear-norm-regularized problem produces a unique, reproducible answer regardless of initialization. The practical difficulty, however, is that the strength of the regularization must be chosen by the analyst. Cross-validation (CV) is the standard remedy, but it is computationally expensive and requires arbitrary choices about the CV scheme.

We resolve both limitations simultaneously through an empirical Bayes framework. By interpreting the nuclear norm penalty as the negative log-prior of a *Matrix Laplace* distribution and placing an uninformative Gamma hyperprior on the regularization strength λ , we obtain a fully Bayesian model in which λ is estimated from the data alongside the latent matrix X . Crucially, a naive maximum a posteriori (MAP) approach fails here due to the Neyman-Scott paradox: collapsing posterior uncertainty into a point estimate causes the effective nuclear norm to be underestimated, biasing λ upward and

over-shrinking the signal. We sidestep this by developing a Coordinate Ascent Variational Inference (CAVI) algorithm that maintains explicit posterior uncertainty over X , leading to well-calibrated automatic regularization without any cross-validation.

2 Characterization of the Matrix Laplace Prior

Let $X \in \mathbb{R}^{m \times n}$ be a random matrix. The Matrix Laplace Prior is a matrix-variate generalization of the standard Laplace (or exponential) distribution, defined by the probability density function:

$$p(X | \lambda) = \frac{1}{Z(\lambda)} \exp(-\lambda \|X\|_*) \quad (1)$$

where $\|X\|_* = \sum_{i=1}^{\min(m,n)} \sigma_i(X)$ is the nuclear norm.

2.1 Marginal Distribution of the Nuclear Norm

Transforming to generalized polar coordinates where the radius is $R = \|X\|_*$, the volume of the nuclear norm ball grows as R^{mn} . The surface area measure therefore scales as R^{mn-1} . Integrating out the directional components, the marginal density of the nuclear norm itself follows a Gamma distribution,

$$\|X\|_* \sim \text{Gamma}(mn, \lambda). \quad (2)$$

This yields the moments $\mathbb{E}[\|X\|_*] = \frac{mn}{\lambda}$ and $\text{Var}(\|X\|_*) = \frac{mn}{\lambda^2}$.

2.2 The Normalizing Constant

Using the Gamma integral, the normalizing constant $Z(\lambda)$ scales with the volume of the unit nuclear norm ball B_* ,

$$Z(\lambda) = \int_{\mathbb{R}^{m \times n}} \exp(-\lambda \|X\|_*) dX = \text{Vol}(B_*) \frac{(mn)!}{\lambda^{mn}} \quad (3)$$

Thus, $Z(\lambda) \propto \lambda^{-mn}$. This explicit scaling is critical for treating λ as a fully Bayesian random variable.

2.3 Joint Density of Singular Values

Assuming $m \geq n$, the change of variables to the Singular Value Decomposition (SVD) yields the joint density of the non-zero singular values $\sigma_1 > \sigma_2 > \dots > \sigma_n > 0$:

$$p(\sigma_1, \dots, \sigma_n) \propto \prod_{i=1}^n \underbrace{\sigma_i^{m-n} e^{-\lambda \sigma_i}}_{\propto \text{Gamma}(\sigma_i, m-n+1, \lambda)} \cdot \underbrace{\prod_{1 \leq i < j \leq n} (\sigma_i^2 - \sigma_j^2)}_{\text{repulsion term}} \quad (4)$$

This characterizes the singular values as a 1D Coulomb gas subject to a logarithmic pairwise repulsion.

3 The Heteroscedastic Low-Rank Model

We consider a setting where we observe a matrix $Y \in \mathbb{R}^{m \times n}$ and a corresponding matrix of known standard errors $S \in \mathbb{R}^{m \times n}$, where s_{ij} is the uncertainty associated with Y_{ij} .

The likelihood of the latent true matrix X is elementwise normal:

$$p(Y | X) = \prod_{i=1}^m \prod_{j=1}^n \mathcal{N}(Y_{ij} | X_{ij}, s_{ij}^2) \quad (5)$$

4 Variational Inference

To avoid the slow convergence of Markov Chain Monte Carlo (MCMC) while preserving the convexity lost by the common $X = UV^T$ factorization, we utilize a variational representation of the nuclear norm. For any matrix X , the nuclear norm can be expressed as a minimization over an auxiliary symmetric positive definite matrix $Q \in \mathbb{S}_{++}^n$,

$$\|X\|_* = \min_{Q \succ 0} \frac{1}{2} \text{Tr}(XQ^{-1}X^T) + \frac{1}{2} \text{Tr}(Q). \quad (6)$$

Three natural choices for the variational family $q(X)$ arise from the symmetric structure of the nuclear norm.

Row factorization. Let $q(X) = \prod_{i=1}^m \mathcal{N}(x_i | \mu_i, \Sigma_i)$, where $x_i \in \mathbb{R}^n$ is the i -th row of X . The expected second moment is

$$\mathbb{E}_q[X^T X] = \sum_{i=1}^m (\mu_i \mu_i^T + \Sigma_i) \in \mathbb{S}_{++}^n, \quad (7)$$

which pairs directly with the $Q \in \mathbb{S}_{++}^n$ auxiliary variable above. When optimizing the Evidence Lower Bound (ELBO), we must evaluate $\mathbb{E}_q[-\lambda \|X\|_*]$. Substituting the variational representation and applying Jensen's inequality gives the tractable lower bound:

$$\mathbb{E}_q[-\lambda \|X\|_*] \geq -\frac{1}{2} \mathbb{E}_q[\lambda] \text{Tr}(\mathbb{E}_q[X^T X] Q^{-1}) - \frac{1}{2} \mathbb{E}_q[\lambda] \text{Tr}(Q). \quad (8)$$

This requires $O(n^3)$ matrix operations per iteration.

Column factorization. We could in principle factorize $q(X)$ over columns instead of rows, but this is equivalent to the row factorization by simply transposing the input data.

5 Coordinate Ascent Variational Inference (CAVI)

We place a Gamma hyperprior on the regularization parameter: $\lambda \sim \text{Gamma}(a_0, b_0)$. We iteratively update the variational distributions to maximize the ELBO.

5.1 Update for Auxiliary Matrix Q

Let $\Psi = \mathbb{E}_q[X^T X] = \sum_{i=1}^m (\mu_i \mu_i^T + \Sigma_i)$. The value of Q that tightly bounds the expectation is the matrix square root of Ψ :

$$Q^* = \Psi^{1/2} = \left(\sum_{i=1}^m \mu_i \mu_i^T + \Sigma_i \right)^{1/2} \quad (9)$$

5.2 Update for Variational Posterior $q(X)$

Given Q and $\bar{\lambda} = \mathbb{E}_q[\lambda]$, the log-posterior for each row x_i is a multivariate Gaussian. Let $S_i = \text{diag}(s_{i1}^2, \dots, s_{in}^2)$ be the known noise variance matrix for row i . The optimal variational updates are:

$$\Sigma_i = \left(S_i^{-1} + \bar{\lambda} Q^{-1} \right)^{-1} \quad (10)$$

$$\mu_i = \Sigma_i \left(S_i^{-1} y_i \right) \quad (11)$$

Notice how the auxiliary matrix Q induces correlation across the columns of X , even if the noise S_i is diagonal.

5.3 Update for Hyperparameter Posterior $q(\lambda)$

Because $Z(\lambda) \propto \lambda^{-mn}$, the log-prior contributes $mn \log \lambda$. Combined with the Gamma prior and the variational bound on the nuclear norm, the optimal $q(\lambda)$ is $\text{Gamma}(a_N, b_N)$. Notably, the scale update simplifies beautifully because $\text{Tr}(\Psi Q^{-1}) + \text{Tr}(Q) = \text{Tr}(\Psi \Psi^{-1/2}) + \text{Tr}(\Psi^{1/2}) = 2\text{Tr}(\Psi^{1/2})$.

$$a_N = a_0 + mn \quad (12)$$

$$b_N = b_0 + \text{Tr}(\Psi^{1/2}) \quad (13)$$

The expected value is updated as $\bar{\lambda} = \frac{a_N}{b_N}$.

5.4 Algorithm

The full procedure is summarized in Algorithm 1. The algorithm requires no cross-validation, naturally discovers the effective rank of the matrix by heavily penalizing negligible principal components, and consists entirely of smooth, strictly convex sub-problems.

Algorithm 1 CAVI for Bayesian Low-Rank Matrix Denoising

Require: Data $Y \in \mathbb{R}^{m \times n}$, Variances $S \in \mathbb{R}^{m \times n}$, Prior a_0, b_0 .

```
1: Initialize  $\mu_i = y_i$  and  $\Sigma_i = I_n$  for  $i = 1, \dots, m$ .
2: Initialize  $\bar{\lambda} = \frac{mn}{\|Y\|_*}$ .
3: repeat
4:   1. Estimate uncentered expected covariance:
5:    $\Psi \leftarrow \sum_{i=1}^m (\mu_i \mu_i^T + \Sigma_i)$ 
6:   2. Update auxiliary matrix:
7:    $Q \leftarrow \Psi^{1/2}$ 
8:   3. Update regularization parameter posterior:
9:    $a_N \leftarrow a_0 + mn$ 
10:   $b_N \leftarrow b_0 + \text{Tr}(Q)$ 
11:   $\bar{\lambda} \leftarrow \frac{a_N}{b_N}$ 
12:  4. Update latent matrix posterior row-by-row:
13:  for  $i = 1, \dots, m$  do
14:     $S_i \leftarrow \text{diag}(s_{i1}^2, \dots, s_{in}^2)$ 
15:     $\Sigma_i \leftarrow (S_i^{-1} + \bar{\lambda} Q^{-1})^{-1}$ 
16:     $\mu_i \leftarrow \Sigma_i S_i^{-1} y_i$ 
17:  end for
18: until ELBO converges
Ensure: Posterior mean  $M = [\mu_1, \dots, \mu_m]^T$ , uncertainties  $\{\Sigma_i\}_{i=1}^m$ , expected penalty  $\bar{\lambda}$ .
```

5.5 Evidence Lower Bound

Convergence is monitored via the Evidence Lower Bound (ELBO), which is also used to select λ in the grid-search mode. Using the variational bound at the optimal $Q = \Psi^{1/2}$ and the normalizing constant $\log Z(\lambda) = -mn \log \lambda + \text{const}$, the ELBO decomposes as

$$\begin{aligned} \mathcal{L} = & \underbrace{-\frac{1}{2} \sum_{(i,j) \in \mathcal{O}} \left[\frac{(y_{ij} - \mu_{ij})^2 + \Sigma_i[j, j]}{s_{ij}^2} + \log(2\pi s_{ij}^2) \right]}_{\text{log-likelihood}} \underbrace{- \bar{\lambda} \text{Tr}(Q) + mn \mathbb{E}_q[\log \lambda]}_{\text{prior on } X \text{ (lower bound)}} \\ & + \underbrace{\frac{1}{2} \sum_{i=1}^m [\log |\Sigma_i| + n(1 + \log 2\pi)]}_{\text{entropy of } q(X)} - \underbrace{\text{KL}(q(\lambda) \parallel p(\lambda))}_{\text{KL for } \lambda} \end{aligned} \quad (14)$$

where $\mathcal{O}$ is the set of observed entries and

$$\text{KL}(\text{Gamma}(a_N, b_N) \parallel \text{Gamma}(a_0, b_0)) = a_N \log b_N - a_0 \log b_0 - \log \Gamma(a_N) + \log \Gamma(a_0) + (a_N - a_0) \psi(a_N) + (15)$$

5.6 Missing Data

The model naturally accommodates missing observations. If entry (i, j) is unobserved, we set $s_{ij} = \infty$, making the noise precision $1/s_{ij}^2 = 0$. The entry then contributes nothing to the likelihood, the row posterior update for μ_i , or the ELBO. Observed and missing entries can be arbitrarily interleaved.

6 Implementation

A reference implementation is provided in the `matlap` Python package using JAX [6] for JIT compilation and automatic differentiation.

Numerical stability. Matrix inverses are never formed explicitly. The matrix square root $Q = \Psi^{1/2}$ is obtained via `jnp.linalg.eigh`, which returns eigenvalues d_k and eigenvectors V so that $Q^{-1} = V \text{diag}(1/d_k) V^\top$ can be applied matrix-free. The per-row precision system $(D_i + \bar{\lambda}Q^{-1})\mu_i = D_i y_i$ is solved with a Cholesky factorisation (`jax.scipy.linalg.cho_factor/cho_solve`).

Vectorisation. The m row updates are independent and are vectorised with `jax.vmap`, giving a single batched Cholesky solve per iteration.

ELBO monotonicity. After updating all rows, Ψ and Q are recomputed from the new $\{\mu_i, \Sigma_i\}$ before the ELBO is evaluated. This ensures the ELBO is computed at a fully self-consistent state and is guaranteed non-decreasing.

Lambda grid search. As an alternative to the empirical-Bayes update, `matlap_grid` accepts a user-supplied grid of λ values, runs CAVI with λ held fixed at each grid point, and returns the solution with the highest ELBO.

7 Scalable Extensions

The full CAVI algorithm stores the m row covariance matrices $\{\Sigma_i\}$ explicitly, requiring $O(mn^2)$ memory—approximately 40 GB for a $10,000 \times 1,000$ matrix in 32-bit precision. We describe two families of extensions that reduce this to $O(mnr)$ or $O(mn)$ while preserving the key algorithmic properties.

7.1 Low-Rank CAVI via the Woodbury Identity

We restrict the variational family to a rank- r *factor subspace*. Instead of maintaining free row covariances $\Sigma_i \in \mathbb{S}_{++}^n$, we fix a shared loading matrix $V_r \in \mathbb{R}^{n \times r}$ with orthonormal columns and parameterise $x_i = V_r z_i$ where $z_i \in \mathbb{R}^r$. The variational posterior is

$$q(z_i) = \mathcal{N}(z_i \mid \hat{z}_i, A_r^{(i)-1}), \quad x_i = V_r z_i, \quad (16)$$

so that the implied row posterior is $\mathcal{N}(x_i \mid \mu_i, \Sigma_i)$ with $\mu_i = V_r \hat{z}_i$ and $\Sigma_i = V_r A_r^{(i)-1} V_r^\top$.

Per-row update. Let $p_i \in \mathbb{R}^n$ be the vector of observed precisions for row i (i.e. $p_{ij} = 1/s_{ij}^2$ for observed entries, 0 for missing). The $r \times r$ factor precision and factor mean are:

$$A_r^{(i)} = \text{diag}(\bar{\lambda}/d_r) + V_r^\top \text{diag}(p_i) V_r, \quad (17)$$

$$\hat{z}_i = A_r^{(i)-1} \left(V_r^\top (p_i \odot y_i) \right), \quad (18)$$

where d_r are the eigenvalues from the current Q update (so $Q^{-1} = V_r \text{diag}(1/d_r) V_r^\top$). Each $A_r^{(i)}$ is $r \times r$, so the Cholesky solve costs $O(nr + r^3)$ per row and $O(mnr)$ over all rows—an n/r -fold reduction from the full $O(mn^3)$.

Q update in factor space. We accumulate the expected second moment restricted to the factor subspace:

$$\Psi_r = \sum_{i=1}^m \left(\hat{z}_i \hat{z}_i^\top + A_r^{(i)-1} \right) \in \mathbb{S}_{++}^r. \quad (19)$$

The eigendecomposition $\Psi_r = W_r \text{diag}(d_r) W_r^\top$ gives the matrix square root $Q_r = W_r \text{diag}(\sqrt{d_r}) W_r^\top$, and the loading matrix is rotated: $V_r \leftarrow V_r W_r$. With this convention the Q update diagonal is d_r , $\text{Tr}(Q) = \sum_k \sqrt{d_{r,k}}$, and the ELBO entropy for row i is $\frac{1}{2} \log |A_r^{(i)-1}| = -\frac{1}{2} \log |A_r^{(i)}|$.

Initialization. V_r is initialized via a randomized SVD (see Section 7.3) of the observed data matrix, projecting Y onto its top- r right singular vectors.

Memory and complexity. The algorithm stores $\hat{z}_i \in \mathbb{R}^r$ and $A_r^{(i)} \in \mathbb{R}^{r \times r}$ per row, giving $O(mr^2 + nr)$ storage instead of $O(mn^2)$. At $m = 10,000$, $n = 1,000$, $r = 50$ this is approximately 44 MB versus 40 GB.

ELBO in factor space. Because x_i lives in the column space of V_r , the entropy of $q(x_i)$ is computed in the r -dimensional factor space: $H[q(z_i)] = \frac{1}{2} \log |A_r^{(i)-1}| + \frac{r}{2} (1 + \log 2\pi)$. The λ update uses $m n$ replaced by $m r$ in the Gamma shape parameter update.

7.2 Low-Rank-Plus-Isotropic CAVI via the Woodbury Identity

The factor-space CAVI restricts Q to be rank- r , assigning zero prior precision to off-subspace directions and forcing all posterior mass onto $\text{col}(V_r)$. We instead restrict Q to a low-rank-plus-isotropic family,

$$Q = V_r \text{diag}(d_r) V_r^\top + \delta (I - V_r V_r^\top), \quad (20)$$

where $\delta > 0$ is an additional *variational* parameter—it has the same status as the in-subspace eigenvalues $d_{r,k}$, representing the off-subspace degrees of freedom of Q and optimized by CAVI alongside them.

Update for δ . With $Q^{-1} = V_r \text{diag}(1/d_r) V_r^\top + \delta^{-1}(I - V_r V_r^\top)$, the nuclear-norm ELBO bound decomposes into independent in-subspace and off-subspace terms. The off-subspace contribution $-\frac{\bar{\lambda}}{2} [\text{Tr}(\Psi_\perp)/\delta + (n - r)\delta]$ is minimised in closed form:

$$\delta^* = \sqrt{\frac{\text{Tr}(\Psi_\perp)}{n - r}}, \quad \text{Tr}(\Psi_\perp) = \text{Tr}(\Psi) - \text{Tr}(\Psi_r), \quad (21)$$

where $\text{Tr}(\Psi) = \sum_i (\|\mu_i\|^2 + \text{Tr}(\Sigma_i))$ and $\text{Tr}(\Psi_r) = \sum_k d_{r,k}^2$. This is the exact analogue of $d_{r,k}^* = \sqrt{[\Psi_r]_{kk}}$ for the in-subspace modes: δ^* is the RMS off-subspace eigenvalue of Ψ .

Posterior updates. Let $\gamma = \bar{\lambda}/\delta$ (off-subspace prior precision), $\tilde{p}_{ij} = p_{ij} + \gamma$, and $c_k = \bar{\lambda}/d_{r,k} - \gamma$. In practice $\delta < d_{r,k}$ (off-subspace modes of Q are smaller than in-subspace), so $c_k < 0$; the posterior precision $\Lambda_i = \text{diag}(\tilde{p}_i) + V_r \text{diag}(c_k) V_r^\top$ is nevertheless PD. The Woodbury identity gives

$$\Sigma_i = \text{diag}(\tilde{p}_i^{-1}) - \text{diag}(\tilde{p}_i^{-1}) V_r \tilde{B}_i^{-1} V_r^\top \text{diag}(\tilde{p}_i^{-1}), \quad (22)$$

$$\tilde{B}_i = \text{diag}(1/c_k) + V_r^\top \text{diag}(\tilde{p}_i^{-1}) V_r \in \mathbb{R}^{r \times r}, \quad (23)$$

where $\tilde{B}_i \succ 0$ by the Schur complement of Λ_i . The posterior mean $\mu_i = \Sigma_i \text{diag}(p_i) y_i$ follows the same structure and has components outside $\text{col}(V_r)$, capturing signal not explained by the current factor subspace. $\text{diag}(\Sigma_i)$ is extracted in $O(nr)$ without forming Σ_i explicitly.

Q and δ updates. $\text{Tr}(\Psi_r)$ and V_r are updated exactly as in Section 7.1. $\text{Tr}(\Psi)$ is accumulated in $O(mn)$ from $\|\mu_i\|^2$ and $\text{diag}(\Sigma_i)$, giving $\text{Tr}(\Psi_\perp)$ and thus $\delta^* = \sqrt{\text{Tr}(\Psi_\perp)/(n-r)}$ and $\gamma \leftarrow \tilde{\lambda}/\delta^*$ in $O(n)$. $\text{Tr}(Q) = \sum_k d_{r,k} + (n-r)\delta^*$.

ELBO. $\log |\Sigma_i| = -\log |\Lambda_i|$ is computed via the matrix determinant lemma applied to Λ_i in $O(n+r^3)$, using all n degrees of freedom rather than only r .

Summary. At the same $O(mnr + mr^3)$ per-iteration cost and $O(mr^2 + mn)$ memory as the factor-space CAVI: (i) μ_i captures off-subspace signal; (ii) the entropy is n -dimensional, correcting the auto- λ over-shrinkage; (iii) no new hyperparameters are introduced— δ is optimized from the data in closed form each iteration.

7.3 Randomized SVD for Approximate Nuclear Norm in SVI

For the gradient-based SVI comparators, the bottleneck is computing $\|X\|_* = \sum_i \sigma_i(X)$ via full SVD at each gradient step, costing $O(mn \min(m, n))$. We replace this with a rank- r randomized SVD approximation:

$$\|X\|_* \approx \sum_{k=1}^r \hat{\sigma}_k(X), \quad (24)$$

where $\hat{\sigma}_1 \geq \dots \geq \hat{\sigma}_r$ are the top- r singular values estimated by power-iteration randomized SVD [5]. The method forms a sketch $Y_\Omega = (XX^\top)^q X\Omega$ for a Gaussian random matrix $\Omega \in \mathbb{R}^{n \times (r+p)}$ with p oversampling columns, then extracts approximate singular values from the resulting small matrix.

To enable gradient-based optimization, we implement the approximation with a `jax.custom_vjp` rule. The forward pass returns $\sum_k \hat{\sigma}_k$; the backward pass propagates the gradient $g \mapsto g \hat{U}_r \hat{V}_r^\top$, which is the standard nuclear norm subgradient restricted to the top- r component and satisfies $\|\nabla\|_{\text{op}} \leq 1$. This reduces the per-step cost from $O(mn^2)$ to $O(mnr)$ for matrices with $n \leq m$.

7.4 Low-Rank Grid Search with Warm-Started Regularisation Path

Rather than estimating λ jointly with X via the empirical-Bayes update, one may instead search a discrete grid of candidate values and select the best by ELBO, as in `matlap_grid`. We combine this with the low-rank CAVI (Section 7.1) to obtain `matlap_grid_lowrank`, which is both scalable and avoids the misspecification of the automatic λ in factor space.

The grid is traversed in *decreasing* order of λ (strong to weak regularisation). The solution at the previous (larger) λ warm-starts the next grid point: the shared loading matrix V_r and per-row factor means $\hat{z}_i$ are reused as the initial state. Because the previous solution is already near the new optimum (the signal grows smoothly as λ decreases), each grid point typically converges in far fewer CAVI iterations than a cold start. After all grid points are evaluated, the λ yielding the highest ELBO (computed in factor space, consistent with Section 7.1) is selected.

This strategy costs $O(G \cdot mn r)$ in total, where G is the number of grid points. On a GPU at $m = 10,000$, $n = 1,000$, $r = 30$, a grid of $G = 7$ points converges in under one second.

7.5 Memory-Efficient Full CAVI via Batched Row Processing

For settings where n is moderate (e.g. $n \lesssim 300$) but m is large, the bottleneck of full CAVI is the simultaneous storage of all m row covariances $\Sigma_i \in \mathbb{S}_{++}^n$. We address this with `matlap_batched`, which processes rows in mini-batches of size B . Each batch computes Σ_i via Cholesky, immediately extracts $\text{diag}(\Sigma_i)$ (needed for the ELBO) and the batch’s contribution to $\Psi = \sum_i (\mu_i \mu_i^\top + \Sigma_i)$, then discards the full Σ_i before moving to the next batch. Peak memory is $O(Bn^2 + mn)$ rather than $O(mn^2)$. With $B = 64$ and $n = 1,000$ this is approximately 256 MB versus 40 GB. The algorithm is otherwise identical to full CAVI (Section ??) and converges to the same solution.

7.6 Scalable Variational Guides for SVI

Two additional guide families reduce the SVI memory footprint to $O(mn)$ while capturing structured posterior correlations:

Shared column-factor guide (`matrix_factor`). A single factor matrix $F \in \mathbb{R}^{n \times k}$ is shared across all m rows:

$$q(x_i) = \mathcal{N}\left(x_i \mid \mu_i, FF^\top + \text{diag}(\psi_i)\right), \quad (25)$$

where $\psi_i \in \mathbb{R}^n$ is a per-row diagonal. This captures shared column-space correlations (as in matrix normal approximations) at $O(mn + nk)$ parameter cost—the same asymptotic order as a fully-factorised guide.

Per-row low-rank guide (`row_lowrank`). Each row i has its own low-rank factor $F_i \in \mathbb{R}^{n \times k}$:

$$q(x_i) = \mathcal{N}\left(x_i \mid \mu_i, F_i F_i^\top + \text{diag}(\psi_i)\right). \quad (26)$$

At rank $k = 5$ this costs approximately 600 MB for $10,000 \times 1,000$, versus 40 GB for full row-MVN covariances.

Both guides are implemented using `numpyro`’s `LowRankMultivariateNormal` distribution and combined with the rSVD nuclear norm approximation, reducing per-step cost to $O(mnr)$.

8 Empirical Comparison

We benchmark all scalable methods on a synthetic matrix completion problem with $m = 10,000$ rows, $n = 1,000$ columns, true rank 15, heteroscedastic Gaussian noise ($\sigma_{ij} \sim \text{Uniform}(0.5, 1.5)$), and 20% of entries held out as a test set. We report test-set RMSE and wall-clock time on a NVIDIA RTX 3090 GPU, averaged over 3 seeds.

Methods. proximal and proximal_cv solve the nuclear-norm penalised least-squares problem via FISTA (300 iterations); λ is set by a heuristic or 5-fold entry-wise CV respectively. matlab_lowrank runs the rank-30 low-rank CAVI with automatic empirical-Bayes λ (80 iterations, converged). matlab_grid_lowrank runs the same low-rank CAVI on a 7-point log-spaced λ grid with warm-started path (converged at each point). matlab_batched runs exact full CAVI with batch size 64, automatic λ (converged; $O(n^3)$ per row, so slow at $n = 1,000$). vi_diagonal runs SVI with a fully-factorised Gaussian guide and full SVD nuclear norm (1000 steps, not converged). vi_diagonal_approx, vi_matrix_factor, and vi_row_lowrank use the rSVD nuclear norm approximation (rank 30) with structured guides (1000 steps, not converged).

Table 1: Test-set RMSE and GPU runtime on a $10,000 \times 1,000$ rank-15 matrix, 3 seeds.

Method	RMSE	Time (s)	Converged	Notes
matlab_grid_lowrank	0.092	0.6	✓	Best overall
proximal_cv	0.105	346	–	500× slower
proximal	0.123	67	–	Fixed λ
matlab_batched	0.153	185	✓	Exact; slow at $n = 1,000$
matlab_lowrank	0.257	1.7	✓	Auto- λ over-shrinks
vi_diagonal	0.207	203	–	1000 steps
vi_diagonal_approx	0.654	52	–	rSVD approx unstable
vi_matrix_factor	0.530	93	–	rSVD approx unstable
vi_row_lowrank	0.529	141	–	rSVD approx unstable

Results.

Discussion. matlab_grid_lowrank achieves the lowest RMSE (0.092) and the shortest runtime (0.6 s), demonstrating that the warm-started regularisation path is both accurate and highly efficient. By evaluating multiple λ values on a shared low-rank basis, it avoids the over-shrinkage of the automatic empirical-Bayes λ in factor space (which inflates the effective regularisation by a factor of $n/r \approx 33$) while remaining far cheaper than the proximal CV baseline.

matlab_batched converges to the exact full-CAVI solution (RMSE 0.153) but is slow at $n = 1,000$ because each row requires an $O(n^3)$ Cholesky; it is intended for settings with moderate n ($\lesssim 300$) where exact row posteriors are needed without the full $O(mn^2)$ memory.

The SVI methods (vi_diagonal, vi_matrix_factor, vi_row_lowrank) have not converged at 1000 steps, and the rSVD nuclear-norm approximation introduces additional gradient noise at rank 30 that degrades the approx variants. With sufficient steps and a higher approximation rank, vi_diagonal achieves competitive RMSE (0.207 at 1000 steps).

References

- [1] D. D. Lee and H. S. Seung. Learning the parts of objects by non-negative matrix factorization. *Nature*, 401(6755):788–791, 1999.
- [2] T. Cai, X. Li, and Z. Ma. Optimal rates of convergence for noisy sparse phase retrieval via thresholded Wirtinger flow. *Annals of Statistics*, 44(5):2221–2251, 2021.

- [3] B. Recht, M. Fazel, and P. A. Parrilo. Guaranteed minimum-rank solutions of linear matrix equations via nuclear norm minimization. *SIAM Review*, 52(3):471–501, 2010.
- [4] E. J. Candès and B. Recht. Exact matrix completion via convex optimization. *Communications of the ACM*, 55(6):111–119, 2012.
- [5] N. Halko, P. G. Martinsson, and J. A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011.
- [6] J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs. <http://github.com/google/jax>, 2018.